

EMCH367 — Homework #2

Jacob C. Vaught, Completed 09-25-2024

Problem 1 (20pt)

True or False

1. Damper element stores the energy.
False: A damper dissipates energy, typically through heat, rather than storing it.
2. Underdamped system has damping ratio between 0 and 1.
True: An underdamped system oscillates with a damping ratio $0 < \zeta < 1$.
3. The stability of a 2nd order system depends on the sign of real part of the poles of the transfer function.
True: Stability is determined by whether the real part of the poles is negative (stable) or positive (unstable).
4. If the step response is stable, then the ramp response is stable.
False: Stability for a step response does not guarantee stability for a ramp input, as they involve different responses.
5. Steady state error is a specific performance indicator.
True: Steady-state error measures the difference between the system output and the desired input in the long term.
6. For a second order system, overshoot always occurs.
False: Overshoot only occurs in underdamped systems; critically damped or overdamped systems do not overshoot.
7. Settling time is always greater than or equal to rise time.
False: Settling time typically includes time for the system to oscillate and dampen, making it longer than the rise time SOMETIMES.
8. Laplace transformation exists for all time functions.
False: The Laplace transform exists only for functions that meet certain conditions, such as being piecewise continuous and of exponential order.
9. Initial value theorem can be used to find steady-state error.
False: The initial value theorem is used to find the initial value of a function, not the steady-state error.
10. Impulse response is the system response with respect to the impulse input.
True: The impulse response represents how a system reacts to a delta function input (impulse).

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
F	T	T	F	T	F	F	F	F	T

Problem #2 (20pt)

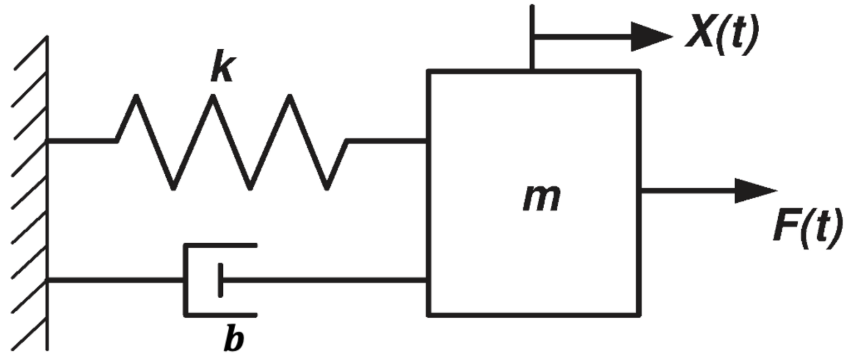


Figure 1: Spring-Mass-Damper System

Consider the spring-mass-damper system shown in Figure 1 with the following parameters:

- Mass, $m = 1$ kg
- Spring constant, $k = 100$ N/m
- Damping coefficient, $b = 8$ N · s/m

(a) Transfer Function Derivation

The equation of motion for the system is given by:

$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = F(t)$$

Taking the Laplace transform of the equation (assuming zero initial conditions):

$$ms^2X(s) + bsX(s) + kX(s) = F(s)$$

Rearranging to solve for the transfer function $G(s) = \frac{X(s)}{F(s)}$:

$$G(s) = \frac{1}{ms^2 + bs + k}$$

(b) Step Response in MATLAB

The MATLAB code to plot the step response for 5 seconds is as follows:

```
% Parameters
m = 1;
b = 8;
k = 100;

% Transfer Function
num = 1;
den = [m b k];
sys = tf(num, den);

% Step Response for 5 seconds
figure;
step(sys, 5);
title('Step Response of the System');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
saveas(gcf, 's1.png');
```

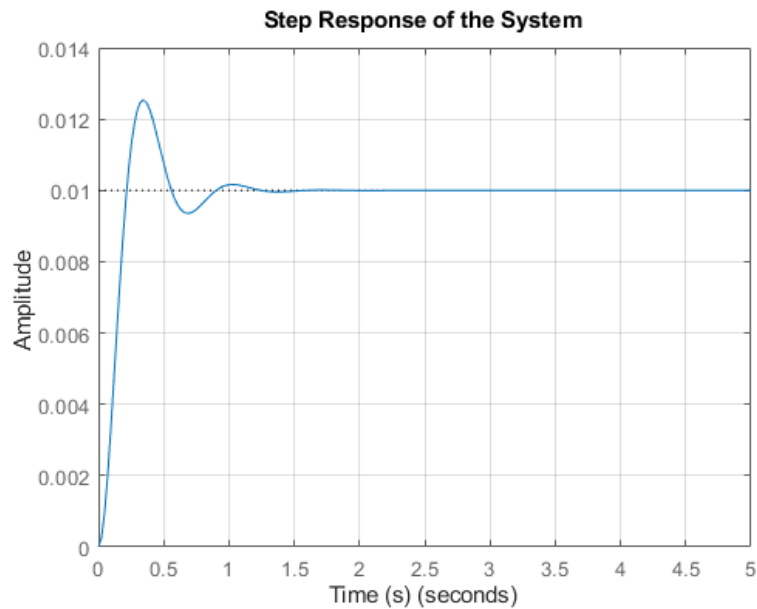


Figure 2: Step Response of the System

(c) Cosine Input Response in MATLAB

The MATLAB code to plot the response to the cosine input $f(t) = \cos(\omega t)$ with $\omega = \frac{\pi}{2} \text{ s}^{-1}$ for 10 seconds is as follows:

```
% Parameters
omega = pi / 2;
t = 0:0.01:10; % Time vector from 0 to 10 seconds

% Cosine Input
f_t = cos(omega * t);

% Response to Cosine Input
[y, t_out] = lsim(sys, f_t, t);

% Plotting
figure;
plot(t, f_t, 'r--', 'DisplayName', 'Input: cos(omega t)');
hold on;
plot(t_out, y, 'b', 'DisplayName', 'System Response');
title('System Response to cos(omega t) Input');
xlabel('Time (s)');
ylabel('Amplitude');
legend show;
grid on;
saveas(gcf, 's2.png');
```

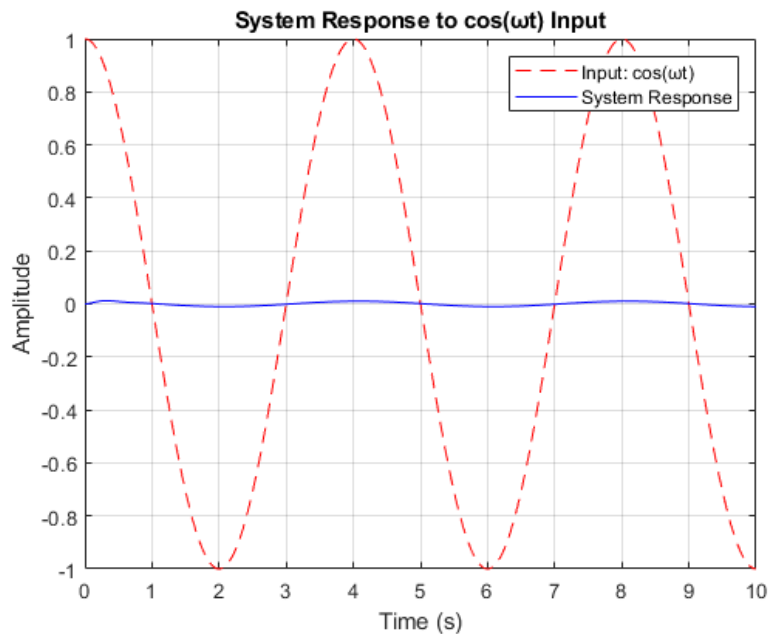


Figure 3: System Response to Cosine Input

(d) Cosine Input Response in SimuLink

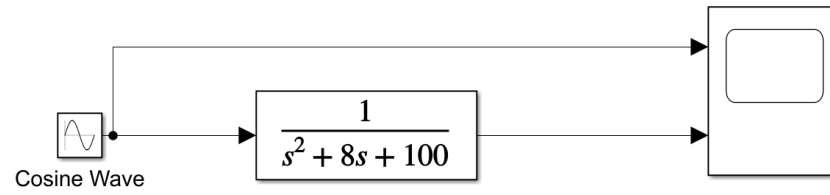


Figure 4: Block System

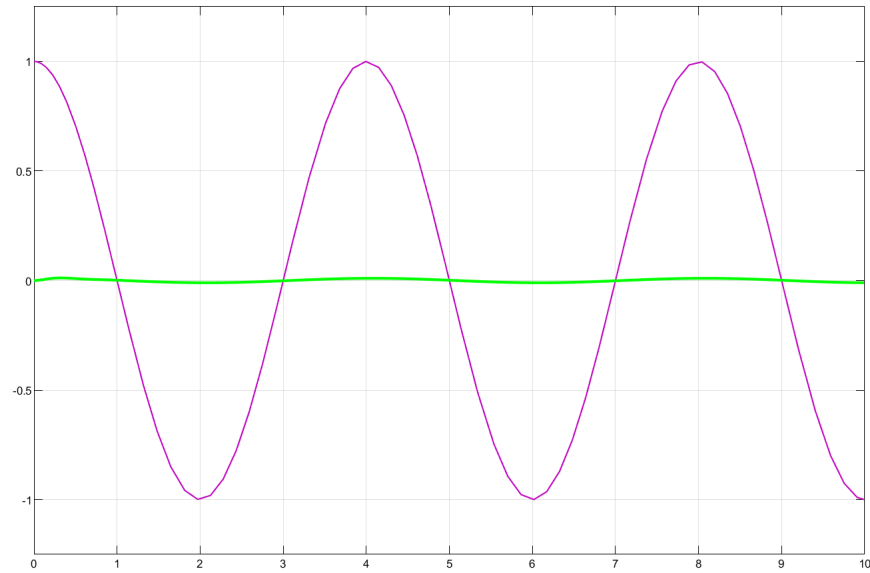


Figure 5: System Response to Cosine Input

Problem 3. (40 pt) Higher Order System

(a) (20 pt)] Consider the system and find the TF of the system $\frac{Y(s)}{U(s)}$ with zero initial conditions.

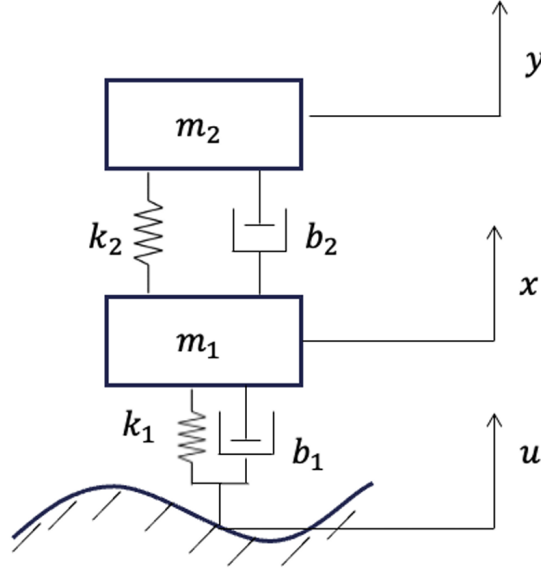


Figure 6: Suspension System

Solving for each mass independently, equations to model each mass can be derived (note that the signs for the mass in question remain the same).

$$m_1 \ddot{x} = k_2(y - x) + b_2(\dot{y} - \dot{x}) + k_1(u - x) + b_1(\dot{u} - \dot{x}) \quad (\text{Eq. 1})$$

$$m_2 \ddot{y} = k_2(x - y) + b_2(\dot{x} - \dot{y}) \quad (\text{Eq. 2})$$

Hence, we can derive the following equations:

$$m_1 \ddot{x} + (b_1 + b_2)\dot{x} + (k_1 + k_2)x = b_2\dot{y} + b_1\dot{u} + k_1u \quad (\text{Eq. 3})$$

$$m_2 \ddot{y} + b_2\dot{y} + k_2y = b_2\dot{x} + k_2x \quad (\text{Eq. 4})$$

Taking the Laplace transform:

$$(m_1 s^2 + (b_1 + b_2)s + (k_1 + k_2))X(s) = b_2 s Y(s) + (b_1 s + k_1)U(s) \quad (\text{Eq. 5})$$

$$(m_2 s^2 + b_2 s + k_2)Y(s) = (b_2 s + k_2)X(s) \quad (\text{Eq. 6})$$

Using Eq. (6) to solve for $X(s)$, we get:

$$X(s) = \frac{m_2 s^2 + b_2 s + k_2}{b_2 s + k_2} Y(s) \quad (\text{Eq. 7})$$

Combining into Eq. (5) to eliminate $X(s)$, we get:

$$[m_1 s^2 + (b_1 + b_2)s + (k_1 + k_2)] \left(\frac{m_2 s^2 + b_2 s + k_2}{b_2 s + k_2} \right) Y(s) = (b_2 s + k_2)Y(s) + (b_1 s + k_1)U(s) \quad (\text{Eq. 8})$$

Simplifying further:

$$\left[\frac{[m_1 s^2 + (b_1 + b_2)s + (k_1 + k_2)] \cdot [m_2 s^2 + b_2 s + k_2]}{(b_2 s + k_2)} \right] Y(s) = (b_2 s + k_2)Y(s) + (b_1 s + k_1)U(s) \quad (\text{Eq. 9})$$

Getting $Y(s)$ on one side:

$$\left[\frac{[m_1 s^2 + (b_1 + b_2)s + (k_1 + k_2)] \cdot [m_2 s^2 + b_2 s + k_2]}{(b_2 s + k_2)} - (b_2 s + k_2) \right] Y(s) = (b_1 s + k_1)U(s)$$

Finally, solving for $\frac{Y(s)}{U(s)}$:

$$\frac{Y(s)}{U(s)} = \frac{(b_1 s + k_1)(b_2 s + k_2)}{(m_1 s^2 + (b_1 + b_2)s + (k_1 + k_2))(m_2 s^2 + b_2 s + k_2) - (b_2 s + k_2)^2} \quad (\text{Eq. 10})$$

(b) (10 pt) Is the system stable?

Due to the system approaching 0 and 1 respectively, the system is stable.

(c) System Stability Analysis + (d) Impulse and Step Responses for 100 Seconds

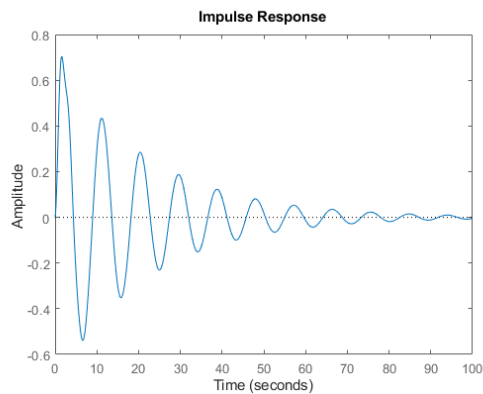


Figure 7: Impulse Response over 100 Seconds

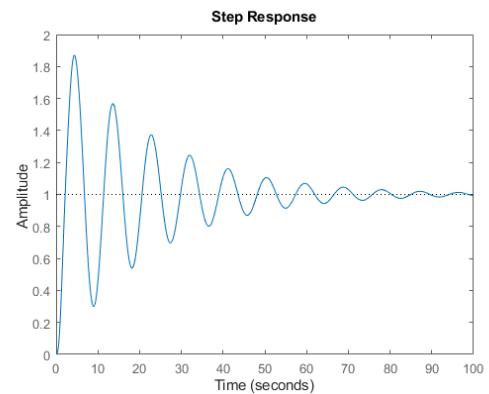


Figure 8: Step Response over 100 Seconds

MATLAB Code for Stability Analysis:

```
% System parameters
m1 = 100;
m2 = 100;
k1 = 100;
k2 = 400;
b1 = 20;
b2 = 50;

% Define the transfer function using symbolic variable s
s = tf('s');
Y_over_U = (b1 * s + k1) * (b2 * s + k2) / ...
    ((m1 * s^2 + (b1 + b2) * s + (k1 + k2)) * (m2 * s^2 + b2 * s + k2) - (b2 * s + k2)^2);

% Time vector for 100 seconds
t = 0:0.1:100;

% Impulse Response
figure;
impz(Y_over_U, t);
title('Impulse Response over 100 Seconds');
xlabel('Time (seconds)');
ylabel('Amplitude');
grid on;
saveas(gcf, 'impulse_response.png');

% Step Response
figure;
step(Y_over_U, t);
title('Step Response over 100 Seconds');
xlabel('Time (seconds)');
ylabel('Amplitude');
grid on;
saveas(gcf, 'step_response.png');
```

Problem #4 (20pt) Performance indicators of 2nd order system.

(a) For step response, find rise time, settling time, overshoot, peak time, and peak value.

Consider a spring-mass-damper system characterized by the differential equation:

$$\ddot{x} + 8\dot{x} + 36x = 36f(t)$$
$$s^2X(s) + 8sX(s) + 36X(s) = 36F(s)$$

For a step input $f(t) = u(t)$, the Laplace transform is:

$$F(s) = \frac{1}{s}$$
$$\frac{X(s)}{F(s)} = \frac{36}{s(s^2 + 8s + 36)}$$

This can be expressed in the standard second-order form:

$$\frac{X(s)}{F(s)} = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

Comparing coefficients, the natural frequency and damping ratio are found to be:

$$\omega_n = \sqrt{36} = 6 \text{ rad/s}$$

$$2\zeta\omega_n = 8 \quad \Rightarrow \quad \zeta = \frac{8}{2 \times 6} = \frac{2}{3}$$

The **peak time** t_p is given by:

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = \frac{\pi}{6 \sqrt{1 - \left(\frac{2}{3}\right)^2}} = \frac{\pi}{4.4724} \approx 0.702 \text{ seconds}$$

The **percentage overshoot** M_p is:

$$M_p = e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100\% = e^{-\frac{\frac{2}{3} \times \pi}{\sqrt{\frac{5}{9}}}} \approx 6.04\%$$

The **settling time** t_s (for 2% tolerance) is:

$$t_s = \frac{4}{\zeta\omega_n} = \frac{4}{\frac{2}{3} \times 6} = 1 \text{ second}$$

The **rise time** t_r can be approximated as:

$$t_r \approx \frac{\pi - \tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)}{\omega_n \sqrt{1 - \zeta^2}} \approx \frac{2.3038}{4.4724} \approx 0.515 \text{ seconds}$$

The **peak value** x_p is given by:

$$x_p = 1 + M_p \approx 1 + 0.0604 = 1.0604$$

The performance indicators are summarized in Table 1.

Performance Indicator	Value
Rise Time (t_r)	$\approx 0.515 \text{ s}$
Peak Time (t_p)	$\approx 0.702 \text{ s}$
Overshoot (M_p)	$\approx 6.04\%$
Settling Time (t_s)	$\approx 1 \text{ s}$
Peak Value (x_p)	≈ 1.0604

Table 1: Performance Indicators of the Second-Order System

(b) Find the steady state error of the ramp response.

To determine the steady-state error (e_{ss}) for a ramp input using the final value theorem, we start with the given system:

$$\frac{X(s)}{F(s)} = \frac{36}{s^2 + 8s + 36}$$

For a ramp input, $f(t) = t$, the Laplace transform is:

$$F(s) = \frac{1}{s^2}$$

We find $X(s)$ by multiplying the transfer function by $F(s)$:

$$X(s) = \frac{X(s)}{F(s)} \cdot F(s) = \frac{36}{s^2 + 8s + 36} \cdot \frac{1}{s^2} = \frac{36}{s^2(s^2 + 8s + 36)}$$

The steady-state error is defined as:

$$e_{ss} = \lim_{t \rightarrow \infty} (f(t) - x(t)) = \lim_{s \rightarrow 0} s [F(s) - X(s)]$$

Substitute $F(s)$ and $X(s)$:

$$e_{ss} = \lim_{s \rightarrow 0} s \left[\frac{1}{s^2} - \frac{36}{s^2(s^2 + 8s + 36)} \right]$$

Simplify the expression:

$$e_{ss} = \lim_{s \rightarrow 0} \left[\frac{1}{s} - \frac{36}{s(s^2 + 8s + 36)} \right]$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s^2 + 8s + 36 - 36}{s(s^2 + 8s + 36)}$$

$$e_{ss} = \lim_{s \rightarrow 0} \frac{s^2 + 8s}{s(s^2 + 8s + 36)} = \lim_{s \rightarrow 0} \frac{s + 8}{s^2 + 8s + 36}$$

Evaluating the limit:

$$e_{ss} = \frac{0 + 8}{0 + 0 + 36} = \frac{8}{36} = \frac{2}{9}$$

Thus, the steady-state error for a ramp input is:

$$e_{ss} = \frac{2}{9}$$

Bonus Problem (20pt)

In Problem 4, find the steady state error of the time response where $f(t) = \sin 6t$. (hint: The steady state error would not converge.)

The steady-state error is defined as:

$$e(t) = f(t) - x(t), \quad (1)$$

where $x(t)$ is the output of the system. Using the Final Value Theorem, the steady-state error is:

$$e_{ss} = \lim_{t \rightarrow \infty} e(t) = \lim_{s \rightarrow 0} sE(s), \quad (2)$$

where $E(s) = F(s) - X(s)$.

The Laplace transform of the input signal $f(t) = \sin(6t)$ is:

$$F(s) = \frac{6}{s^2 + 36}. \quad (3)$$

The output $X(s)$ is given by:

$$X(s) = \frac{36}{s^2 + 8s + 36} \cdot \frac{6}{s^2 + 36} = \frac{216}{(s^2 + 8s + 36)(s^2 + 36)}. \quad (4)$$

The error signal in the Laplace domain is:

$$E(s) = F(s) - X(s) = \frac{6}{s^2 + 36} - \frac{216}{(s^2 + 8s + 36)(s^2 + 36)}. \quad (5)$$

For the steady-state error to converge, the limit $\lim_{s \rightarrow 0} sE(s)$ must exist. However, due to the periodic nature of the sinusoidal input, the response $x(t)$ will also oscillate indefinitely, causing $e(t)$ to remain oscillatory. **Thus, the steady-state error does not converge.**

MATLAB Verification

To verify this behavior, we simulate the time-domain response using MATLAB. The following code generates the time response of the system and visualizes the error:

```
numerator = 36;
denominator = [1, 8, 36];
sys = tf(numerator, denominator);
t = 0:0.01:10; % Time vector from 0 to 10 seconds
input_signal = sin(6*t)'; % Input: sin(6t) transposed
output_signal = lsim(sys, input_signal, t);
```

```
figure;
subplot(2, 1, 1);
plot(t, input_signal, 'b', 'LineWidth', 1.5);
title('Input Signal: sin(6t)');
xlabel('Time (s)');
ylabel('Amplitude');
```

```
grid on;
subplot(2, 1, 2);
plot(t, output_signal, 'r', 'LineWidth', 1.5);
title('System Response');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
```

```
error_signal = input_signal - output_signal;
figure;
plot(t, error_signal, 'k', 'LineWidth', 1.5);
title('Error Signal: f(t) - x(t)');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
```

The plots generated by the MATLAB script confirm that the error signal $e(t) = f(t) - x(t)$ oscillates indefinitely and does not converge to a constant value.

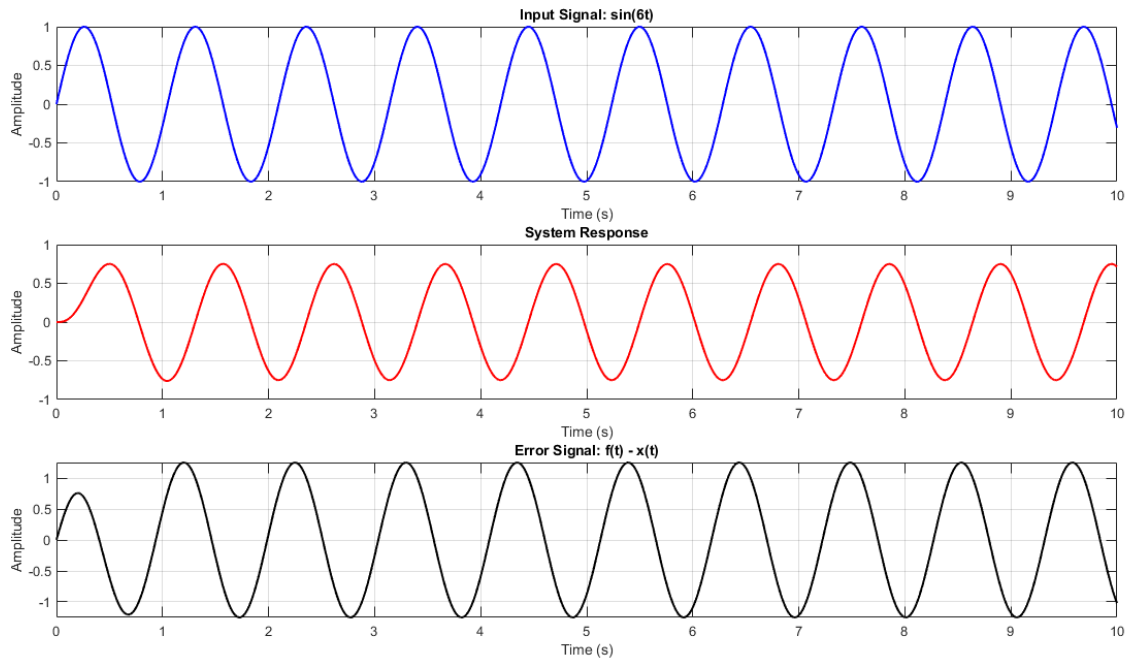


Figure 9: Time response of the system and the error signal for the input $\sin(6t)$.